

Homework — 1.1.3 Input, Output and Storage Devices — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [3] — 1 mark each: (a) **barcode scanner/reader** (accept camera-based scanner); (b) **screen/monitor** (accept touch screen); (c) **(receipt) printer**.
2. [2] — 1 mark: data is represented as a pattern of **pits and lands** burned into the disc surface (a spiral track). 1 mark: a **laser** is shone at the surface and the **reflected light** differs between pits and lands, which is interpreted as the binary data.
3. [3] — Advantages (1 mark each, max 2): faster read/write; no moving parts → more durable/shock-resistant; silent; lower power consumption; lighter/more compact. Disadvantage (1 mark): higher cost per GB; finite number of write cycles; (typically lower maximum capacity than HDD at the same price).
4. [3] — 1 mark recommendation: **magnetic tape** (accept HDD for cost reasons). Up to 2 marks justification linked to the scenario: very **low cost per GB / cheap for huge capacity**; suited to **archive/backup** that is **rarely accessed** (serial access is acceptable when random access is not needed); long shelf life / removable for off-site storage. (*Reject generic pros not linked to "cheap rarely-accessed archive"; cap at recommendation mark.*)
5. [2] — 1 mark: RAM is **volatile** (contents lost when power is removed) and holds the **programs and data currently in use**. 1 mark: ROM is **non-volatile** (contents kept without power) and stores **firmware such as the BIOS/boot program**; its contents cannot normally be changed.
6. [1] — Virtual storage = storage that appears local to the user but is physically held **remotely** (e.g. on a cloud provider's/third party's servers) and accessed **over the internet/a network**.

Part B (6 marks)

7. [2] — 1 mark per difference (max 2): RISC few simple fixed-length instructions vs CISC many complex variable-length; complexity in software/compiler (RISC) vs hardware (CISC); RISC instructions typically one cycle vs CISC several; RISC easier to pipeline; RISC lower power.
8. [2] — 1 mark: a GPU has a **very large number of (simpler) cores**. 1 mark: so it applies the **same operation to many items of data in parallel**, giving far higher throughput than processing them one at a time.
9. [2] — 1 mark: **Memory Data Register**. 1 mark: holds the **data or instruction** just fetched from memory, or about to be written to memory.

Total: 14 + 6 = 20